LEADERBOARD ARCHITECTURE SPECIFICATION

Normalized Demographic Scoring & Multiplier Framework

1. CORE ARCHITECTURE OVERVIEW

The system segregates standings dynamically based on the member's designated `fitness_goal`. Members operate exclusively within their chosen tracks to maintain situational relevance. To ensure fairness across diverse body styles, age cohorts, and biological sexes, the engine overlays raw algorithmic tracking values with a scientifically verified demographic multiplier curve.

Key Structural Parameters:

- **Partitioning:** Independent dashboard pipelines run for Fat Loss, Muscle Gain, and Maintenance.
- **Sorting Logic:** High-to-low positioning based entirely on the `final_leaderboard_score`.
- **Multi-tier Limits:** Fat Loss and Muscle Gain normalization targets feature a maximum scaling cap of `2.00`. The Maintenance pipeline maintains a rigid operational ceiling of exactly `100`.

2. MASTER DEMOGRAPHIC MULTIPLIER MATRIX

The system engine queries the database birthdate and gender records to automatically isolate the correct multiplier from the matrix below before pushing calculation objects into ranking arrays.

Age Bracket	Male Multiplier	Female Multiplier	Physiological Grounding & System Buffers
Under 18	0.95	1.06	Adolescent growth-surge metabolic correction handicap.
18 – 29	**1.00 (Base)**	**1.12**	**Standard population control baseline. Accounts for female essential lipid bounds.**
30 – 35	1.04	1.16	BMR deceleration inflection phase (Calculated ~1.3% linear decay).
36 – 40	1.09	1.22	Sarcopenia protection boundary factor.
41 – 45	1.15	1.29	Endocrine drift correction and structural fat retention offset.
46 – 50	1.22	1.37	Advanced metabolic slowdown progression parameter.
51 – 55	1.30	1.46	Proteolytic synthesis drop mitigation index.
56 – 60	1.39	1.56	Musculoskeletal tissue resistance preservation scaling.
61+	1.50	1.68	Maximum compensation setting for advanced age homeostasis variance.

3. FINALIZED LEADERBOARD NORMALIZED FORMULAS

Track 1: Fat Loss Leaderboard (`fitness_goal = 'fat_loss'`)

Measures demographic-weighted percentage improvements across Body Fat Percentage (BF%) intervals.

$$\textit{Raw Fat Loss Score} = ((\textit{Baseline BF\%} - \textit{Current BF\%}) \div \textit{Baseline BF\%}) \times 100$$

$$\textit{Final Leaderboard Score} = \textit{Raw Fat Loss Score} \times \textit{Multiplier}$$

EXECUTION EXAMPLE

Member Profile: Female, 38 years old (Multiplier = 1.22)
Assessments: Baseline BF% = 35% | Current BF% = 28%
Raw Output: ((35 - 28) ÷ 35) × 100 = 20.00
Final Calculated Score: 20.00 × 1.22 = **24.40 Points**

Track 2: Muscle Gain Leaderboard (`fitness_goal = 'muscle_gain'`)

Tracks relative progression loops focusing entirely on absolute Lean Body Mass (LBM) increments.

$$\text{Raw Muscle Gain Score} = ((\text{Current LBM} - \text{Baseline LBM}) \div \text{Baseline LBM}) \times 100$$

$$\text{Final Leaderboard Score} = \text{Raw Muscle Gain Score} \times \text{Multiplier}$$

EXECUTION EXAMPLE

Member Profile: Male, 52 years old (Multiplier = 1.30)

Assessments: Baseline LBM = 60 kg | Current LBM = 66 kg

Raw Output: ((66 - 60) ÷ 60) × 100 = 10.00

Final Calculated Score: 10.00 × 1.30 = **13.00 Points**

Track 3: Maintenance Leaderboard (`fitness_goal = 'maintenance'`)

Scores homeostatic retention. To maintain a strict system ceiling of 100, the demographic multiplier inversely impacts the cumulative metric variance (the deviation penalty) rather than inflating the final output points directly.

$$\text{Raw Variance (Penalty)} = \text{ABS}(\text{Current BF\%} - \text{Baseline BF\%}) + \text{ABS}(\text{Current LBM} - \text{Baseline LBM})$$

$$\text{Final Leaderboard Score} = 100 - (\text{Raw Variance} \times (2 - \text{Multiplier}))$$

EXECUTION EXAMPLE

Member Profile: Female, 48 years old (Multiplier = 1.37 → Penalty Buffer = 0.63)

Assessments: Baseline BF% = 18%, Current BF% = 19% | Baseline LBM = 65kg, Current LBM = 64kg

Raw Variance Calculation: ABS(19 - 18) + ABS(64 - 65) = 1 + 1 = 2.00

Final Calculated Score: 100 - (2.00 × 0.63) = 100 - 1.26 = **98.74 Points**

4. DATABASE SCHEMA UPGRADES

To support execution loops within data warehouses, map these precise parameters to your structural backend objects:

- `baseline_bf_percent` (DECIMAL) / `current_bf_percent` (DECIMAL)
- `baseline_lbm` (DECIMAL) / `current_lbm` (DECIMAL)
- `member_age` (INTEGER) / `member_sex` (VARCHAR)
- `demographic_multiplier` (DECIMAL) - *Evaluated via lookup rules.*
- `final_leaderboard_score` (DECIMAL) - *Core indexed execution column for ranking loops.*

5. BACKEND ENGINE SCRIPT IMPLEMENTATION

```
/**
 * GymCatalyst Engine Pipeline - Normalization Scoring
 */
function processMemberLeaderboardScore(member) {
  let rawScore = 0;
  let finalScore = 0;

  if (member.fitness_goal === 'fat_loss') {
    rawScore = ((member.baseline_bf - member.current_bf) / member.baseline_bf) * 100;
    finalScore = rawScore * member.demographic_multiplier;

  } else if (member.fitness_goal === 'muscle_gain') {
    rawScore = ((member.current_lbm - member.baseline_lbm) / member.baseline_lbm) * 100;
    finalScore = rawScore * member.demographic_multiplier;

  } else if (member.fitness_goal === 'maintenance') {
    let bfVariance = Math.abs(member.current_bf - member.baseline_bf);
    let lbmVariance = Math.abs(member.current_lbm - member.baseline_lbm);
    let totalVariance = bfVariance + lbmVariance;

    // Apply inverse multiplier buffer to protect scores against metabolic biological drift
    let bufferFactor = 2.0 - member.demographic_multiplier;
    finalScore = 100.0 - (totalVariance * bufferFactor);
  }

  // Bind to object storage rounded cleanly for database efficiency
  return parseFloat(finalScore.toFixed(2));
}
```